higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MULTIMEDIA SERVICES
NQF LEVEL 4

(10040184)

8 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 10 pages.

NCV

TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.15) in the ANSWER BOOK.

1.1.1 Apart from <b> tag, what other tag makes text bold?

A <fat>
B
C <black>
D <emp>

1.1.2 What type of language is HTML?

A Scripting language
B Mark-up language
C Programming language
D Network protocol

1.1.3 An attempt to make a computer resource unavailable to its intended users is called …

A denial-of-service attack.
B virus attack.
C worms attack.
D botnet process.

1.1.4 Which of the following is not a browser?

A Microsoft Bing
B Netscape Navigator
C Mozilla Firefox
D Opera

1.1.5 When a DNS server accepts and uses incorrect information from a host that has no authority to give that information, it is called …

A DNS lookup.
B DNS hijacking.
C DNS spoofing.
D DNS lockup.

1.1.6 WPA2 is used for security in …

A ethernet.
B bluetooth.
C WiFi.
D optic fibre.

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1.1.7 Spyware infects a computer through …

A printing.
B theft of personal info.
C recording web info.
D B and C

1.1.8 How would you be able to protect yourself from phishing?

A Call your local fish store.
B Get an antivirus program.
C Change passwords every 4–6 months.
D B and C

1.1.9 How does identity theft cause problems for online users?

A Steals social security
B Locates where you live and obtains all your information
C Steals bank information
D All the abovementioned

1.1.10 What characteristic makes the internet so attractive?

A The 'secure' surroundings within which it is implemented
B The ability to provide an open, easy-to-use network
C It eliminates the need for firewalls
D You don't require a fast computer to use the internet

1.1.11 … is to protect data and passwords.

A Encryption
B Authentication
C Authorisation
D Nonrepudiation

1.1.12 A firewall can protect against …

A viruses.
B unauthenticated interactive logins from the 'outside' world.
C fire.
D connecting to and from the 'outside' world.

1.1.13 The main purpose of access control is to …

A authorise full access to authorised users.
B limit the actions or operations that a legitimate user can perform.
C stop unauthorised users accessing resources.
D protect computers from viral infections.

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1.1.14 Which of the following is NOT a good property of a firewall?

A Only authorised traffic must be allowed to pass through it.
B The firewall itself should be immune to penetration.
C It should allow for easy modification by authorised users.
D Traffic must only be allowed to pass from inside to outside the firewall.

1.1.15 This technique tries to take information from the user, by getting the user to actively give the information. Common things taken are credit cards, user names, passwords and home addresses.

A Phishing
B Scam
C Spyware
D Fishing

(15 × 1) (15)

1.2 Give ONE word or term for each of the following descriptions. Write only the word or term next to the question number (1.2.1–1.2.15) in the ANSWER BOOK.

1.2.1 A database model in which the data is organised in a tree like structure.

1.2.2 The collection of data stored in a standardised format, designed to be shared by multiple users

1.2.3 Encoding data with a key value so the data becomes unreadable

1.2.4 Collection of data for one class or entity. It consists of columns for each attribute and a row of data for each specific entity or object.

1.2.5 This person is a specialist who is trained in the administration of a particular DBMS. He/she is trained in the details of installing, configuring and operating the DBMS.

1.2.6 A data field which has the purpose to uniquely identify a record. It helps to enforce data integrity and avoid duplication.

1.2.7 A database structure that helps speed up database queries for large data sets

1.2.8 This process ensures that only authorised users can access a database or certain aspects of a database

1.2.9 A database management which system information is represented in the form of objects. It is different from a relational database which is table-oriented.

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1.2.10 Software that manages all aspects of working with a database, from storing and securing the data to enforcing data integrity rules, to providing forms for data entry and manipulation

1.2.11 A type of backup that copies all the data that has changed since the last full backup

1.2.12 The computer display that allows the users to interact with the database

1.2.13 This must match an existing primary key in another table

1.2.14 A database in which portions of the database are stored in multiple physical locations and processing is spread among multiple database nodes

1.2.15 The most basic and complete type of backup operation. This type of backup makes a copy of all data to another set of media, which can be a tape, an external HDD or a DVD or CD.

(15 × 1) (15)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.15) in the ANSWER BOOK.

1.3.1 The critical layer is a part of the TCP/IP protocols and its connection to the internet.

1.3.2 The first browser created by Microsoft was called Cello.

1.3.3 A cookie is a small text file that a web server may store on a user's computer for the purpose of uniquely identifying the user during future visits to the website.

1.3.4 A digital subscriber line (DSL) is an internet connection that works over a regular telephone line and allows the user to be on the phone and on the internet at the same time.

1.3.5 HTTPS is a version of HTTP which is not secure and provides communication using SSL.

1.3.6 Java applet is a Java program that can run within a webpage and can be downloaded to the user's computer from a webserver.

1.3.7 Spyware refers to hidden software with the purpose of gathering data about a user's web surfing habits.

1.3.8 A software tool used to search the *world wide web* for web pages and indexing them accordingly is called a web spider.

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1.3.9 Pop 3 is an e-mail protocol that allows you to retrieve e-mails from the server.

1.3.10 A landing page is the page where a visitor sees when he/she leaves a website.

1.3.11 A liquid layout is one that is based on percentages of the browser window's size.

1.3.12 The information contained in the meta-data is viewable on the web page.

1.3.13 Plug-ins provides a way to extend the functionality of a website without having to redo the core coding of the site.

1.3.14 Dynamic IP is an IP address that does not change every time when a device is connected to the internet.

1.3.15 Flash movies cannot be loaded on websites because the files are too big.

(15 × 1) (15)

1.4 Give the full term for the following abbreviations. Write only the term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

1.4.1 ISDN

1.4.2 ADSL

1.4.3 IIS

1.4.4 DNS

1.4.5 HTTPs

(5 × 1) (5)
[50]

TOTAL SECTION A: 50

SECTION B

QUESTION 2: NETWORKS AND SERVERS

2.1 You discover that your junior network technician has connected a client's computer with a cable that is longer than specified.

Highlight the problems she might face when connecting the clients computer to the network. (3)

2.2 Describe TWO major drawbacks of implementing a ring topology as a network solution. (2)

2.3 Briefly describe why it has become necessary for the internet community to implement IPv6 instead of IPv4. (3)

2.4 On opening the packaging of a new switch, you discover that it is labelled as being 10/100 switch.

Justify the labelling of the switch. (4)

2.5 You have been as by the network administrator to join two subnets (LAN networks) to form one large network. He has provided you with a bridge.

Determine the role of the network bridge and where it will be placed on the network. (8)

[20]

QUESTION 3: DATABASE MANAGEMENT

3.1 Produce the SQL statement to create a database called *Hospital.* (3)

3.2 Produce the SQL statement to create the table below called *Patients.* (3)

3.3 Study the table below and answer the questions that follow:

PatientsID	Surname	Address	Fees	CellNo
9203078123	Noah	1 Brits City	R300-00	0786623
8809236671	Mutsi	12 Durban	R2909-00	0835524
2008128792	Ntali	9 Midrand	R2300-40	0118865
2010106672	Anna	23A Pretoria	R2708-00	0125564
7407248777	Botsinai	67 Natal Road	R2301-00	0318854

3.3.1 Produce the code that will list all the patients whose fees are more than R2 300-00. (4)

3.3.2 Identify the Primary Key in this table and also suggest the best alternate Primary Key for this Table. (2)

3.3.3 Produce the SQL statement to display all the records in this table. (4)

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3.3.4 Produce the SQL statement to DELETE the record for Ntali from this table. (4)
[20]

QUESTION 4: WEB PLANNING

4.1 Identify SIX examples of bad web page design that you will avoid when designing web pages. (6)

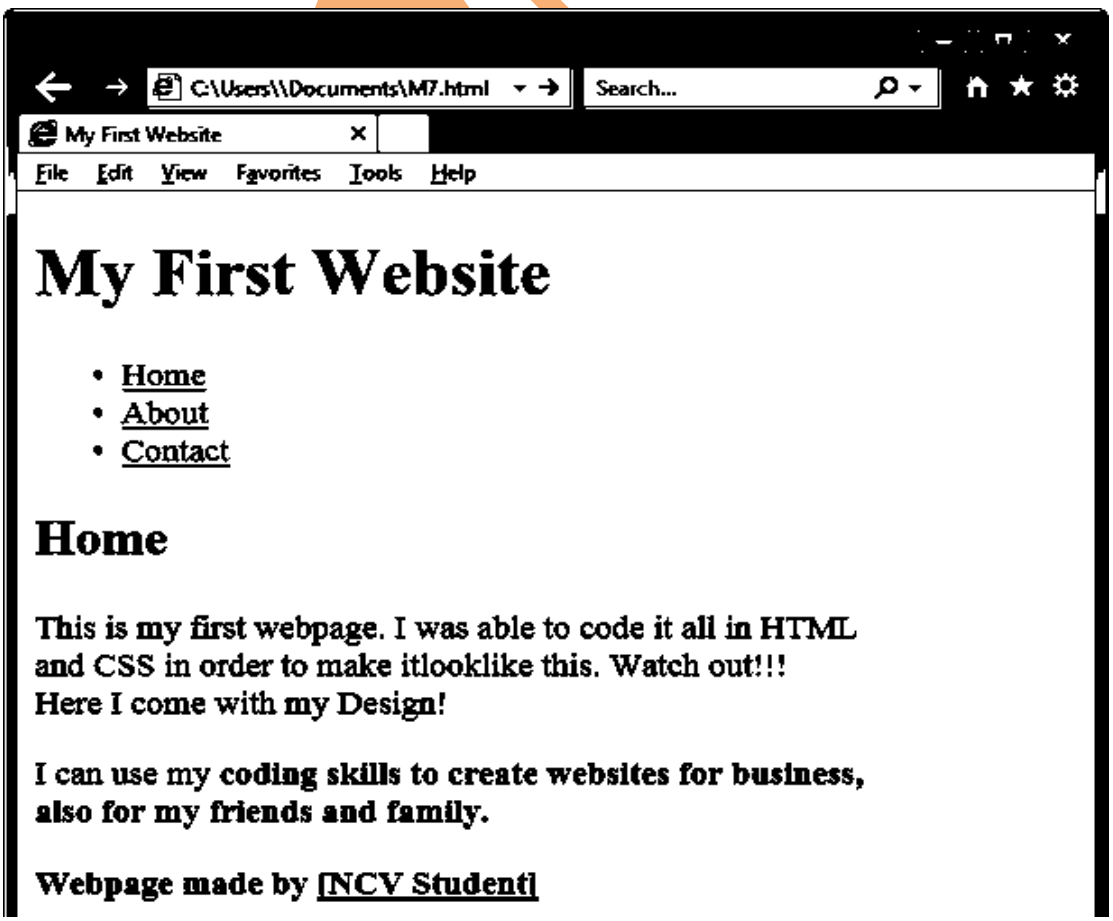
4.2 Demonstrate by the use of an example the purpose of the *marquee tag* in the HTML code. (6)

4.3 State THREE benefits or advantages of using a ring network topology (3 × 2) (6)

4.4 State the very first browser to be created for the internet. (2)
[20]

QUESTION 5: WEB DESIGN

Study the webpage below and answer the questions that follow:



5.1 Produce the HTML code for this website. (10)

5.2 Produce the JavaScript code to display the heading 'Welcome to my Home Page'. (3)

5.3 Design a comment 'I like website Programming' using JavaScript. (2)

5.4 You are required to link this website to an external CSS file called nicestyles.css.

Produce the piece of code needed to link the web page to the external CSS file. (5)
[20]

QUESTION 6: INTERNET SECURITY

6.1 Discuss THREE reasons why people create computer viruses and malware. (3 × 2) (6)

6.2 Tabulate the FOUR major differences between a *firewall* and an *antivirus* program. (8)

6.3 Briefly explain the purpose of an intrusion detection system. (3)

6.4 Discuss where you would place an intrusion prevention system in a network. (3)
[20]

TOTAL SECTION B: 100
GRAND TOTAL: 150

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